

CARLOS JORQUES, Ph.D.

Senior Control Software Engineer | Technical Leader

Location: Valencia, Spain

Phone: +34689707376

E-mail: carlosjorques@gmail.com

Senior control and embedded software engineer with 12+ years of automotive R&D experience across electric, combustion, and autonomous vehicle systems. Specialized in real-time embedded control development, with strong experience in controller architecture, diagnostics, fault handling, and full-cycle validation. Current work includes ISO 26262 implementation activities, with focus on system requirements, traceability, reviews, and engineering alignment. Previous roles include software leadership, Scrum Master responsibilities, Automotive SPICE process adoption, model-based development strategy, validation coordination, and stakeholder communication across Scania, TRATON, Volvo CE, and Mahle.

CORE COMPETENCES

- **Automotive Process and Safety:** ISO 26262 implementation support, Automotive SPICE / ASPICE adoption support, safety-related software, model quality, process discipline
- **Tooling and Software Lifecycle:** Jira, Git, GitLab, Jenkins, SVN, Perforce Helix, CI/CD, MATLAB/Simulink, Embedded Coder, AUTOSAR workflows
- **Model-Based Development:** MATLAB/Simulink, Embedded Coder, auto-code workflows, control-oriented modeling
- **Embedded Software Development:** C, C++, Python, AUTOSAR, diagnostics, fault handling, ECU integration, calibration, Git, CI/CD, requirements analysis, data analysis, debugging
- **Verification and Validation:** MIL, SIL, HIL, bench, wind tunnel, test track, vehicle testing, logged-data analysis, debugging, release support
- **Collaboration:** technical leadership, mentoring, Agile/SAFe, supplier coordination, international teamwork

Languages: English (proficient), Spanish (native), Catalan (native), Swedish (proficient)

WORK EXPERIENCE

Alten Delivery Center | Valencia, Spain | Apr 2023 – Present

Senior Control Software Engineer | Technical Leader

Senior consultant and technical leader for embedded control software deliveries, leading 3 engineers across requirements, architecture, validation, and customer delivery.

TRATON / Scania: Thermal management and electrified powertrain controls

- Designed real-time control architecture and predictive thermal control in MATLAB/Simulink, from concept evaluation to ECU-ready implementation in C++ within Adaptive AUTOSAR.
- Developed model-based embedded software for electric truck thermal management, covering batteries, power electronics, chargers, pumps, valves, fans, and auxiliaries.
- Supported ISO 26262 ASIL-B software development in a GitLab CI/CD environment using SAFe.

Volvo CE: Autonomous electric hauler propulsion control

- Developed and integrated traction-control software for autonomous electric haulers operating in harsh off-road conditions.
- Designed propulsion torque-control architecture for dual motors and individual axle control, including torque allocation with redundant actuators.
- Implemented diagnostics, monitoring, unit/integration tests, and system-level validation aligned with functional safety requirements.
- Validated software through SIL, HIL, and vehicle testing in manual and autonomous modes.

Mahle Electronics | Valencia, Spain | Jan 2022 – Mar 2023

Senior Development Software Engineer

- Led software development for an e-bike motor control project using model-based development and calibration workflows.
- Designed and calibrated a virtual PMSM temperature model with 98% model accuracy.
- Developed application-layer software for a wireless charger and DC/DC converter.
- Defined model-based development strategy for an AUTOSAR-based platform on Infineon AURIX.
- Contributed to Automotive SPICE adoption and served as Scrum Master.

Senior Development Software Engineer

- Designed control architecture for thermal management in electric powertrains and implemented control strategies for electrified vehicle applications.
- Developed estimation methods for battery SOC, SOP, and SOH in mild-hybrid systems.
- Improved thermal-management efficiency by 2% through control-function development, calibration, and validation in test bench and vehicle environments.
- Performed end-to-end validation through MIL, SIL, HIL, wind tunnel, test track, and on-road activities.

Industrial PhD Researcher

- Researched in-cycle closed-loop combustion control algorithms for diesel engines running with biofuels.
- Investigated, designed, and implemented advanced closed-loop control algorithms for heavy-duty engines, translating high-speed sensing and predictive models into real-time control actions implemented on an FPGA.
- Analyzed hardware-resource constraints for real-time controller implementation with efficiency gains up to +1.8%unit.
- Published 15 scientific papers, generated 4 patents, and mentored thesis students in industrial-academic collaboration.

Development Software Engineer

- Developed embedded software for diagnostics, thermal management, oil-system control, and safety-related functions using MATLAB/Simulink and C code generation.
- Contributed from design and simulation through implementation, testing, calibration, debugging, and flashing.

Other early experience: Scania, Master Thesis Student, Sodertälje (2014); **Arduino Verkstad**, Hardware Engineer, Malmö (2013); **Sony Mobile Communications**, Software Researcher, Lund (2013)

EDUCATION

- **Industrial Ph.D. Mechanical Engineering** – Scania and Lund University, Sweden | 2021
Thesis on the design, implementation, and optimization of real-time closed-loop control, estimation, predictive models, and embedded execution for automotive applications.
- **M.Sc. Electrical Engineering** – Lund University, Sweden | 2014
Automatic Control and Automation: Specialization in control systems, embedded software, and mechatronics.
- **M.Sc. Industrial Engineering** – Polytechnic University of Valencia, Spain | 2014
Electronics and Automation: Specialization in automation, control theory, and energy systems.

Courses and Training

- LangChain and LLM with Python, Alten, 2026
- Project Management: Product Owner + Kanban, Cualtis, 2024
- Power Electronic Converters, 2022
- Introduction to Electromobility, Swedish Electromobility Centre, 2020
- Scania Innovation Challenge, Scania, 2019
- GT-Power, Gamma Technologies, 2015
- Control of Mobile Robots, Coursera, 2013

ACHIEVEMENTS

- **Patents:** 5 patents in control, estimation, diagnostics, optimization, and real-time implementation
- **Publications:** 15 scientific articles in real-time control, predictive control, virtual sensing, combustion diagnostics, stochastic optimization, and FPGA implementation.
- **Recognitions:** Outstanding Oral Presentation Award, SAE 2017; academic honours and competitive scholarships from Lund University, UPV, COIICV, and TIME Association.

Selected Publications

- *FPGA Implementation of In-Cycle Closed-Loop Combustion Control Methods*, SAE Technical Paper, 2021.
- *Modular Design and Integration of In-Cycle Closed-Loop Combustion Controllers for a Wide Range of Operating Conditions*, 2021 American Control Conference (ACC), 2021
- *In-Cycle Closed-Loop Combustion Control for Pilot Misfire Compensation*, SAE Int. J. Advances and Current Practices in Mobility, 2021.
- *Bayesian Method for Fuel Mass Estimation of Short Pilot Injections Based on its Misfire Probability*, American Control Conference, 2020.